1.

What is the atomic number of an element equal to?
A. Number of neutrons
B. Number of protons
C. Number of electrons and neutrons combined
D. Atomic mass

Answer: B

2.

Which subatomic particle has a negative charge?
A. Proton
B. Neutron
C. Electron
D. Nucleus

Answer: C

3.

Which scientist proposed the planetary model of the atom?
A. Dalton
B. Thomson
C. Rutherford
D. Bohr

Answer: D

4.

What is the mass number of an atom with 11 protons and 12 neutrons?
A. 11
B. 12
C. 23
D. 24

Answer: C

5.

Which element is located in Group 17 of the periodic table?
A. Neon

B. Sodium
C. Chlorine
D. Magnesium

Answer: C

6.

What type of bond involves the sharing of electrons?
A. Ionic
B. Metallic
C. Covalent
D. Hydrogen

Answer: C

7.

Which compound is ionic?
A. CO_2
B. H_2O
C. NaCl
D. CH_4

Answer: C

8.

What is the electron configuration of sodium (Na)?
A. 2-8-1
B. 2-8-2
C. 2-7-1
D. 2-8-8-1

Answer: A

9.

Which particle determines the identity of an element?
A. Neutron
B. Electron
C. Proton
D. Isotope

Answer: C

10.

What is the pH of a neutral solution at 25°C?
A. 0
B. 7
C. 10
D. 14

Answer: B

11.

Which state of matter has a definite volume but no definite shape?
A. Solid
B. Liquid
C. Gas
D. Plasma

Answer: B

12.

Which law states that matter cannot be created or destroyed?
A. Boyle’s Law
B. Charles’s Law
C. Law of Conservation of Mass
D. Law of Multiple Proportions

Answer: C

13.

What is the molar mass of H2O?
A. 16 g/mol
B. 18 g/mol
C. 20 g/mol
D. 22 g/mol

Answer: B

14.

Which gas is most abundant in Earth’s atmosphere?
A. Oxygen
B. Carbon dioxide

C. Nitrogen
D. Hydrogen

Answer: C

15.

What happens to gas pressure if volume decreases while temperature stays constant?
A. Pressure decreases
B. Pressure increases
C. Pressure stays the same
D. Pressure becomes zero